

IUB DRONE COMMERCIAL AUTONOMY PLATFORM

OPERATOR MANUAL, SYSTEM INSTALLATION & NETWORK SETUP GUIDE

SUAS 2026 Tactical Ground Control Station & Autonomous Vision Suite

1. System Architecture Overview

The **IUB Drone Commercial Autonomy Platform** is an aerospace-grade companion computer system designed for autonomous flight, vision perception, and Ground Control Station (GCS) telemetry operations. It connects onboard Jetson AI hardware directly to Matek ArduPilot flight controllers over high-speed serial MAVLink links.

Component	Hardware Model / Protocol	Function & Specifications
Flight Controller	Matek H743-WING V2 (ArduPilot)	400Hz MAVLink control over /dev/ttyTHS1 @ 921600 baud.
Companion Computer	NVIDIA Jetson Orin Nano / Nano	Ubuntu 22.04 LTS, MAXN 15W mode, PyTorch FP16 CUDA.
Primary Camera	SIYI A8 Mini 4K Gimbal Camera	HD RTSP stream (rtsp://192.168.144.25:8554/main.264) + USB fallback.
Avoidance/VIO Cam	Intel RealSense D455 3D	6-DOF VIO optical flow & 3D depth perception (/dev/video4).
LiDAR Altimeter	TFmini-S Micro LiDAR	Millimeter-precision altitude readout (/dev/ttyUSB0 @ 115200 baud).
Ground Station	Web GCS Asynchronous Engine	Zero-dependency dark-mode web workstation on port 8080.

2. Installation & Software Environment Setup

Follow these step-by-step commands to install and optimize the Jetson companion computer environment:

```
# 1. Clone the repository on Jetson companion computer
cd ~
git clone https://github.com/chishti055nsu/SUASS_DRONE.git IUB_DRONE
cd ~/IUB_DRONE

# 2. Grant serial port permissions for Matek FC and TFmini LiDAR
sudo usermod -a -G dialout,video $USER
sudo chmod 666 /dev/ttyTHS1 /dev/ttyUSB0 /dev/video* 2>/dev/null || true

# 3. Engage Max Hardware Performance (CPU & CUDA GPU Clock Lock)
bash scripts/maximize_jetson_performance.sh
```

3. Ground Station to Drone Network Connection Setup

To establish high-speed wireless connectivity between the Ground Station computer (laptop/tablet) and the drone companion computer:

- IP Configuration:** Connect Ground Station computer and Jetson to the SIYI HM30 Datalink / WiFi router. Set Jetson Static IP to **192.168.144.100** and Ground Station IP to **192.168.144.150**.
- Ping Verification:** Run `ping 192.168.144.100` from Ground Station to verify sub-millisecond network link.

- **RTSP Stream:** SIYI A8 Mini 4K stream broadcasts at `rtsp://192.168.144.25:8554/main.264`. If network RTSP is offline, system automatically falls back to USB video capture (`/dev/video0`).

4. Web GCS Control Center Operation Manual

Start the Web GCS server on the Jetson computer:

```
bash scripts/start_web_gcs.sh
```

Access the interface from your Ground Station browser at **`http://192.168.144.100:8080`**.

Web GCS Tab	Primary Controls & Features
Top Telemetry Header	Real-time 400Hz MAVLink status, 18 SAT RTK GPS, 25.2V 6S Power readout, and clickable ARM / DISARM badge .
Tab 1: Tactical Dashboard	2D ENU Radar with Click-to-Fly (GOTO), Horizon/Compass PFD, ■ DISARM MOTORS NOW, START MISSION, 5.5KG HEAVY-LIFT , and Dynamic Hover Throttle Calibrator .
Tab 2: SIYI A8 Mini Feed	Live HD RTSP feed with FP16 YOLOv8 object detection, Target-Payload Matching table (Mannequin -> Water Bottle), and Gemma AI Scene Reasoning.
Tab 3: RealSense D455	3D VIO optical flow tracking, 3-Sector Obstacle Avoidance clearance (Left/Center/Right), and TFmini-S LiDAR altitude indicator.
Tab 4: Virtual Remote	Touch/Mouse Teleop Nudge Controls and Precision Throttle Override slider.

5. Step-by-Step Flight Operations & Emergency Procedures

Pre-Flight Checklist:

1. Check 6S LiPo battery voltage ($\geq 24.0V$).
2. Verify MAVLink 400Hz connection and 3D RTK GPS Fix (≥ 14 Satellites).
3. Verify TFmini-S LiDAR altitude readout and SIYI A8 Mini camera feed.

Emergency Motor Cut-Off Procedure:

EMERGENCY DISARM PROCEDURE:

*If a flight anomaly occurs, click **■ DISARM MOTORS NOW** on the Web GCS control panel or click the top header status badge **ARMED - LIVE**.*

Triple-Redundant Safety Action Triggered:

1. Direct MAVLink force-disarm command (`param1=0, param2=21196` force magic) sent over serial.
2. Throttle PWM locked to 1000 across all channels.
3. ROS 2 MAVROS arming service call (`/mavros/cmd/arming value: false`) dispatched.

Motors cut power instantly!

6. Quick Troubleshooting Guide

Symptom	Root Cause	Resolution
Web GCS page not loading	Server not started or port blocked.	Run 'bash scripts/start_web_gcs.sh' on Jetson. Check IP.
SIYI A8 Mini stream black	RTSP IP mismatch or cable unplugged.	System auto-falls back to USB video (/dev/video0). Check Ethernet IP.
Tab switch lag	Dual streams downloading simultaneously.	Updated software automatically pauses hidden stream on tab switch.
Motors not disarming	Serial connection lost.	Click '■ DISARM MOTORS NOW' or header badge for triple-redundant disarm.